

2.4. Possible choices of Δ

The class of possible differences in trends Δ must be specified by the researcher, and the choice of Δ will depend on the economic context. We highlight several possible choices of Δ that may be reasonable in empirical applications and formalize intuitive arguments that are commonly made by applied researchers regarding possible violations of parallel trends. Throughout our discussion, we write $\delta_{pre} = (\delta_{-T}, \dots, \delta_{-1})'$ and $\delta_{post} = (\delta_1, \dots, \delta_{\bar{T}})'$, with δ_0 normalized to zero. This aligns the notation with Example 1, where δ corresponds to the difference in trends between treated and comparison groups, and δ_0 is normalized to zero.

2.4.1. Bounding relative magnitudes. In empirical applications, researchers may be willing to assume that the confounding factors which produce non-parallel trends in the post-treatment periods are not too much larger in magnitude than the confounding factors in the pre-treatment periods. In their empirical application to right-to-carry gun laws, [Manski and Pepper \(2018\)](#) operationalize this intuition by calibrating bounds on $|\delta_1|$ to the largest violations of parallel trends in the pre-treatment period (see their Table 3).⁶ Such a restriction can be formalized in our framework by imposing that $\delta \in \Delta^{RM}(\bar{M})$ for $\bar{M} \geq 0$, where

$$\Delta^{RM}(\bar{M}) = \{\delta : \forall t \geq 0, |\delta_{t+1} - \delta_t| \leq \bar{M} \cdot \max_{s < 0} |\delta_{s+1} - \delta_s|\}.$$

$\Delta^{RM}(\bar{M})$ bounds the maximum post-treatment violation of parallel trends between consecutive periods by $\bar{M}$ times the maximum pre-treatment violation of parallel trends. We use the abbreviation RM for “relative magnitudes.” The choice $\Delta^{RM}(\bar{M})$ may be reasonable if the researcher suspects that possible violations of parallel trends are driven by confounding economic shocks that are of a similar magnitude to confounding economics shocks in the pre-period. When the number of pre-treatment and post-treatment periods is similar, a natural benchmark may be $\bar{M} = 1$, which bounds the worst-case post-treatment difference in trends by the equivalent maximum in the pre-treatment period.⁷

2.4.2. Example: non-staggered DiD (continued). In the three-period DiD model ($T = \bar{T} = 1$), assuming $\delta \in \Delta^{RM}(\bar{M}) = \{(\delta_{-1}, \delta_1)' : |\delta_1| \leq \bar{M}|\delta_{-1}|\}$ bounds the magnitude of δ_1 based on the magnitude of δ_{-1} . The larger the magnitude of the pre-treatment violation in parallel trends, $|\delta_{-1}|$, the wider the range of possible post-treatment violations of parallel trends. ▲

2.4.3. Smoothness restrictions. In other empirical settings, researchers may be worried about confounding from secular trends (*e.g.* long-run changes in labour supply) that they suspect evolve smoothly over time. In such settings, it is common for empirical researchers to control for a linear group-specific time trend.⁸ This approach is valid if the difference in trends is linear,

6. In their application, [Manski and Pepper \(2018\)](#) observe the outcome for the entire population of interest, and thus their observed pre-treatment data corresponds with δ_{pre} rather than $\hat{\beta}_{pre}$.

7. In settings where later pre-periods are thought to be more informative, the researcher could use a different value of $\bar{M}$ for different pre-periods, *e.g.* imposing that $|\delta_{t+1} - \delta_t| \leq \max\{|\delta_0 - \delta_{-1}|, 2 \cdot |\delta_{-1} - \delta_{-2}|\}$. Likewise, although we’ve considered bounds on changes across consecutive periods, the researcher could also impose bounds on changes across multiple periods, *e.g.* $|\delta_{t+2} - \delta_t| \leq \bar{M} \cdot \max_{s < -1} |\delta_{s+2} - \delta_s|$.

8. Specifically, researchers often augment specification (2) with group-specific linear trends, an approach [Dobkin et al. \(2018\)](#) refer to as a “parametric event-study.” An analogous approach is to estimate a linear trend using only observations prior to treatment, and then subtract out the estimated linear trend from the observations after treatment ([Bhuller et al., 2013](#); [Goodman-Bacon, 2018, 2021](#)).